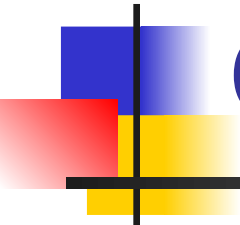
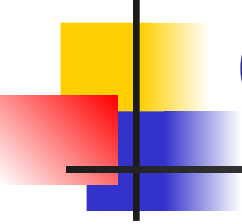


Ch5 Mining Frequent Patterns, Associations, and Correlations

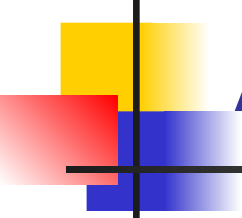


Dr. Bernard Chen Ph.D.
Troy University



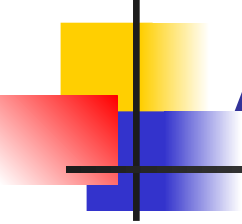
Outline

- Association Rules
- Misleading Rules
- Multi-level Association Rules



What Is Frequent Pattern Analysis?

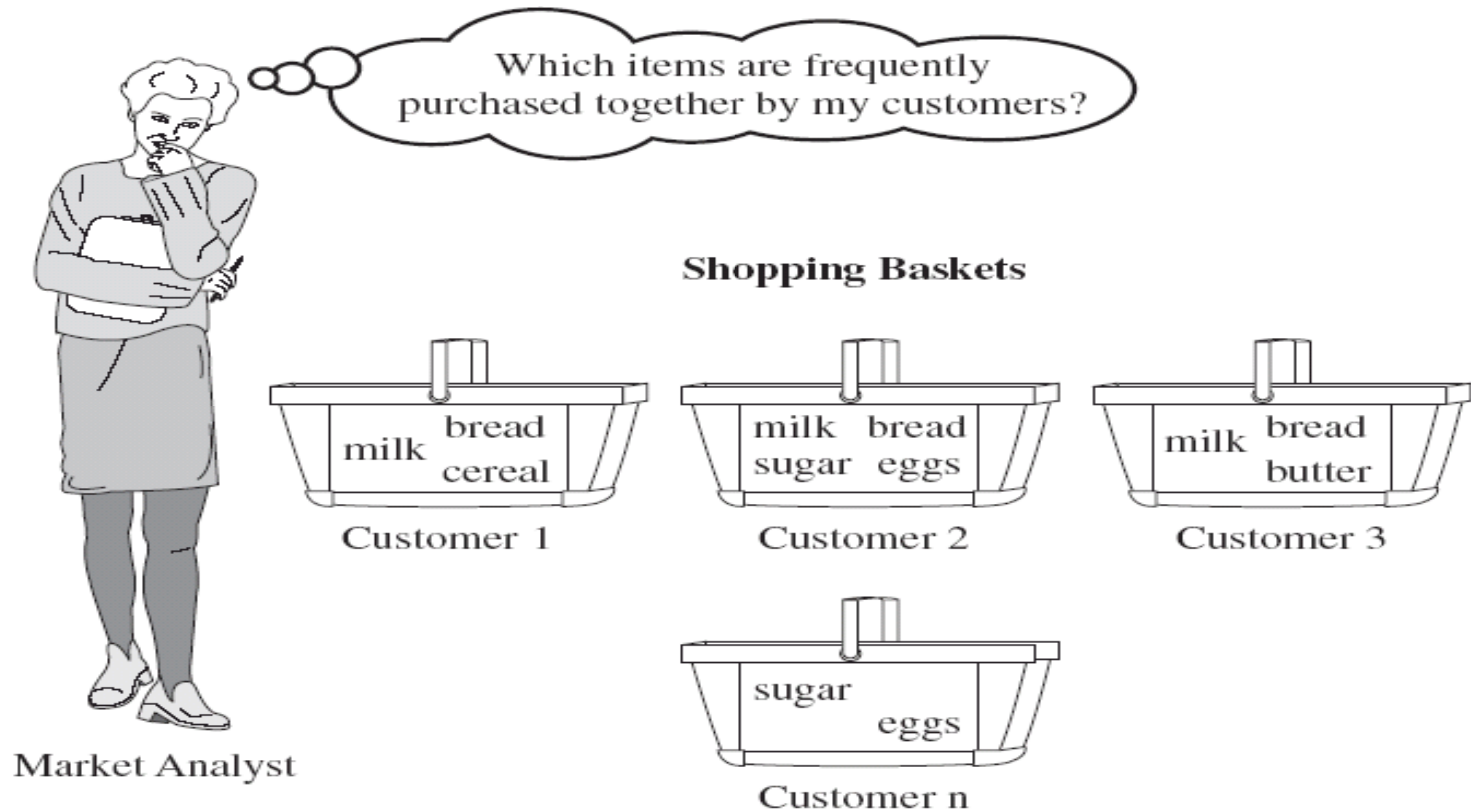
- **Frequent pattern**: a pattern (a set of items, subsequences, substructures, etc.) that occurs frequently in a data set
- First proposed by Agrawal, Imielinski, and Swami [AIS93] in the context of **frequent itemsets** and **association rule mining**

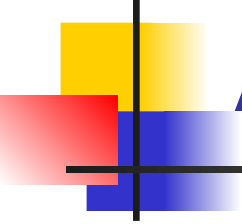


What Is Frequent Pattern Analysis?

- Motivation: Finding inherent regularities in data
 - What products were often purchased together? bread and milk?
 - What are the subsequent purchases after buying a PC?
 - What kinds of DNA are sensitive to this new drug?
 - Can we automatically classify web documents?
- Applications
 - Basket data analysis, cross-marketing, catalog design, sale campaign analysis, Web log (click stream) analysis, and DNA sequence analysis.

Association Rules





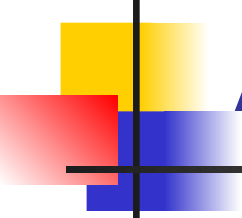
Association Rules

- **support**, s , **probability** that a transaction contains $X \cup Y$

$$\text{Support}(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$$

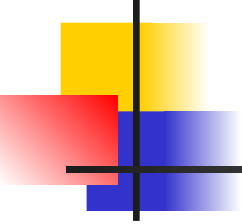
- **confidence**, c , **conditional probability** that a transaction having X also contains Y

$$\text{Confidence}(X \Rightarrow Y) = \frac{|X \cup Y|}{|X|}$$



Association Rules

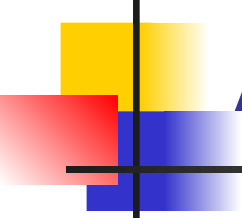
- Then user need to define:
 - Minimum Support
 - Minimum Confidence



Let us focus on SUPPORT

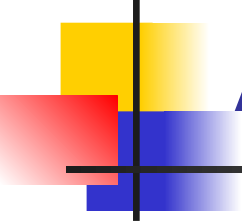
- **support**, s , **probability** that a transaction contains $X \cup Y$

$$\text{Support}(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$$



Association Rules

- Let's have an example
 - T100 1,2,5
 - T200 2,4
 - T300 2,3
 - T400 1,2,4
 - T500 1,3
 - T600 2,3
 - T700 1,3
 - T800 1,2,3,5
 - T900 1,2,3
- Suppose our **minimum support** is 2/9

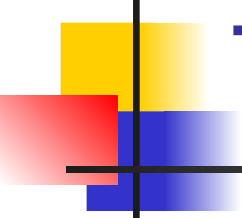


Association Rules

$$\text{Support}(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$$

- Let's have an example
 - T100 1,2,5
 - T200 2,4
 - T300 2,3
 - T400 1,2,4
 - T500 1,3
 - T600 2,3
 - T700 1,3
 - T800 1,2,3,5
 - T900 1,2,3

- $|T|=9$



To calculate support

$$\text{Support}(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$$

- Since the grocery has only 5 items, the basic way is to try all items:

- TWO item set

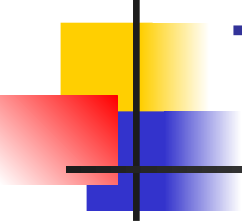
- (1,2) (1,3) (1,4) (1,5)
- (2,3) (2,4) (2,5)
- (3,4) (3,5)
- (4,5)

Support (1,2) = 4/9 (pass)

Support (2,3) = 4/9 (pass)

Support (3,5) = 1/9

■ T100	1,2,5
■ T200	2,4
■ T300	2,3
■ T400	1,2,4
■ T500	1,3
■ T600	2,3
■ T700	1,3
■ T800	1,2,3,5
■ T900	1,2,3



To calculate support

$$\text{Support}(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$$

- Since the grocery has only 5 items, the basic way is to try all items:

- THREE item set

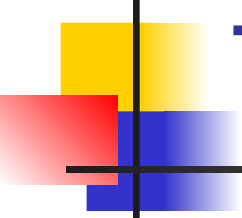
- (1,2,3) (1,2,4) (1,2,5) (1,3,4) (1,3,5) (1,4,5)
- (2,3,4) (2,3,5) (2,4,5)
- (3,4,5)

Support (1,2,3) = 2/9 (pass)

Support (2,3,4) = 0/9

Support (3,4,5) = 0/9

■ T100	1,2,5
■ T200	2,4
■ T300	2,3
■ T400	1,2,4
■ T500	1,3
■ T600	2,3
■ T700	1,3
■ T800	1,2,3,5
■ T900	1,2,3



To calculate support

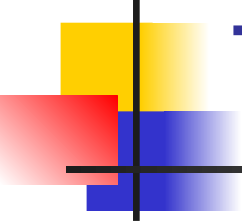
$$\text{Support}(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$$

- Since the grocery has only 5 items, the basic way is to try all items:
- FOUR item set
 - (1,2,3,4) (1,2,3,5)
 - (2,3,4,5)

Support (1,2,3,4) = 0/9

Support (2,3,4,5) = 0/9

■ T100	1,2,5
■ T200	2,4
■ T300	2,3
■ T400	1,2,4
■ T500	1,3
■ T600	2,3
■ T700	1,3
■ T800	1,2,3,5
■ T900	1,2,3



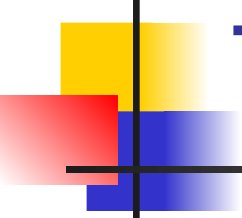
To calculate support

$$\text{Support}(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$$

- Since the grocery has only 5 items, the basic way is to try all items:
- FIVE item set
 - (1,2,3,4,5)

Support (1,2,3,4,5) = 0/9

■ T100	1,2,5
■ T200	2,4
■ T300	2,3
■ T400	1,2,4
■ T500	1,3
■ T600	2,3
■ T700	1,3
■ T800	1,2,3,5
■ T900	1,2,3



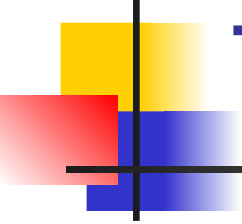
To calculate support

$$\text{Support}(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$$

- This is the situation based on our store has only **5** items
- In real grocery store like Kroger, they have **thousands** of items

- Is there a better way?

■ T100	1,2,5
■ T200	2,4
■ T300	2,3
■ T400	1,2,4
■ T500	1,3
■ T600	2,3
■ T700	1,3
■ T800	1,2,3,5
■ T900	1,2,3



To calculate support

$$\text{Support}(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$$

- Of course, it's called the **Apriori**
- It proceeds by identifying the **frequent** individual items (the item that passes the minimum support) in the database and **extending** them to larger and larger item sets as long as those item sets appear sufficiently often in the database.

To calculate support $Support(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$

- Start with one item set

Scan D for count of each candidate

→

C_1	
Itemset	Sup. count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

Compare candidate support count with minimum support count

→

L_1	
Itemset	Sup. count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

ALL Pass

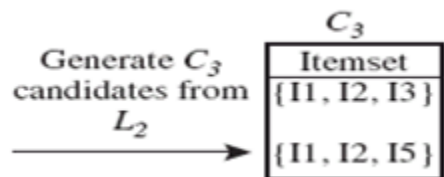
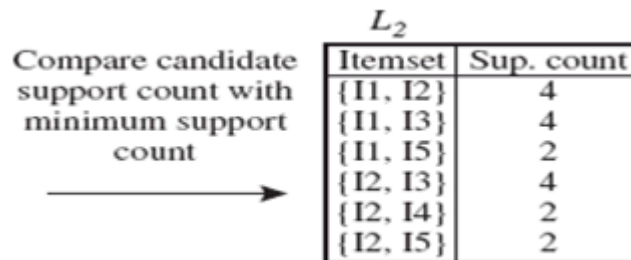
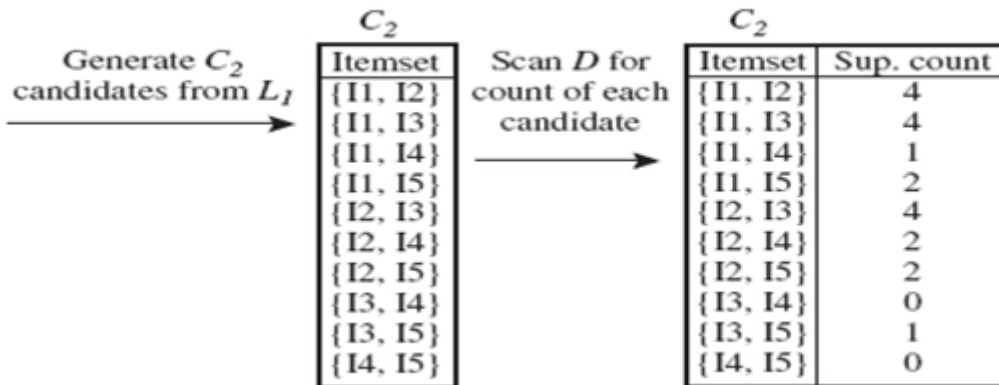
Generate C_2 candidates from L_1

→

C_2	
Itemset	
{I1, I2}	
{I1, I3}	
{I1, I4}	
{I1, I5}	
{I2, I3}	
{I2, I4}	
{I2, I5}	
{I3, I4}	
{I3, I5}	
{I4, I5}	

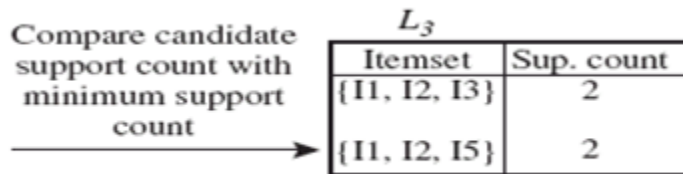
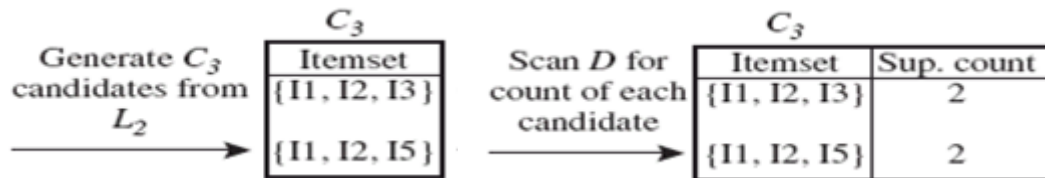
■ T100	1,2,5
■ T200	2,4
■ T300	2,3
■ T400	1,2,4
■ T500	1,3
■ T600	2,3
■ T700	1,3
■ T800	1,2,3,5
■ T900	1,2,3

To calculate support $Support(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$



- T100 1,2,5
- T200 2,4
- T300 2,3
- T400 1,2,4
- T500 1,3
- T600 2,3
- T700 1,3
- T800 1,2,3,5
- T900 1,2,3

To calculate support $Support(X \Rightarrow Y) = \frac{|X \cup Y|}{|T|}$



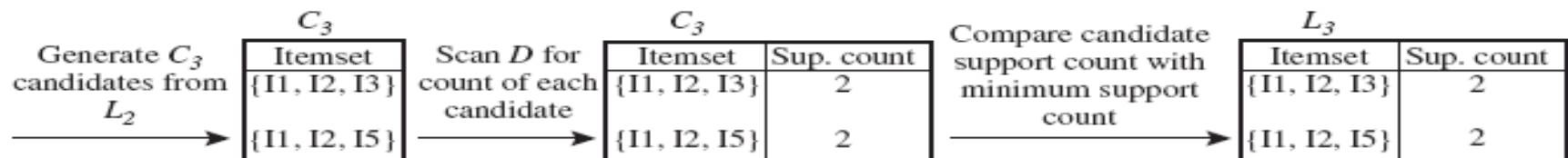
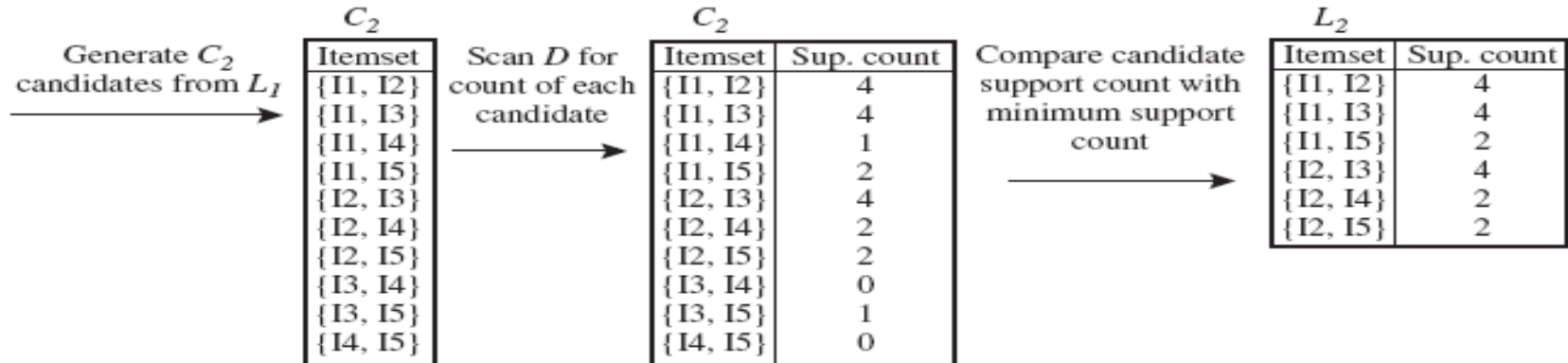
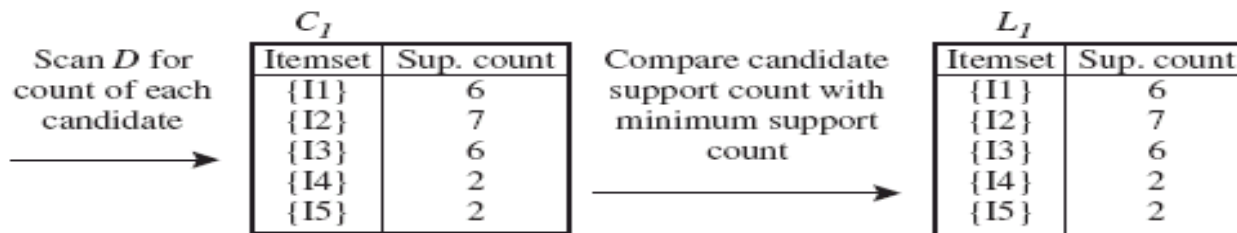
- $\Rightarrow \{I1, I2, I3, I5\}$
- $Support \{I1, I2, I3, I5\} = 1/9$
- No Five item set can be generated

- T100 1,2,5
- T200 2,4
- T300 2,3
- T400 1,2,4
- T500 1,3
- T600 2,3
- T700 1,3
- T800 1,2,3,5
- T900 1,2,3

Association Rules with Apriori

Minimum support=2/9

Minimum confidence=60%



The Apriori Algorithm

- Pseudo-code:

C_k : Candidate itemset of size k

L_k : frequent itemset of size k

$L_1 = \{\text{frequent items}\};$

for ($k = 1; L_k \neq \emptyset; k++$) **do begin**

C_{k+1} = candidates generated from L_k ;

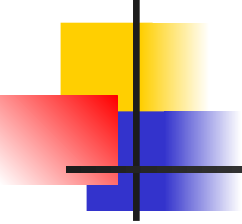
for each transaction t in database **do**

increment the count of all candidates in C_{k+1}
that are contained in t

L_{k+1} = candidates in C_{k+1} with min_support

end

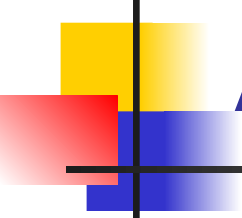
return $\cup_k L_k$;



Exercise

- A dataset has five transactions, let min-support=60%
- Find all frequent item sets using Apriori

TID	Items_bought
T1	M, O, N, K, E,
T2	Y
T3	D, O, N, K , E,
T4	Y
T5	M, A, K, E
	M, U, C, K ,Y
	C, O, O, K, I ,E



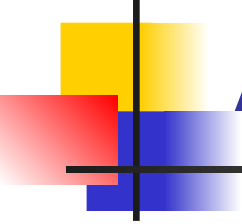
Association Rules

- **support**, s , **probability** that a transaction contains $X \cup Y$

$$\text{Support}(X \Rightarrow Y) = \frac{\text{Precedent } |X \cup Y|}{\text{Consequent } |T|}$$

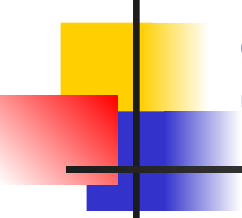
- **confidence**, c , **conditional probability** that a transaction having X also contains Y

$$\text{Confidence}(X \Rightarrow Y) = \frac{\text{Precedent } |X \cup Y|}{\text{Consequent } |X|}$$



Association Rules

- Now, Let us take a look of **CONFIDENCE**
 - Suppose our **minimum confidence: 60%**
- **confidence**, c , **conditional probability** that a transaction contains Y
$$Confidence(X \Rightarrow Y) = \frac{|X \cup Y|}{|X|}$$

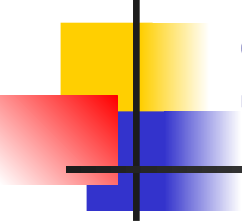


Strong Association Rule

- **Strong association rules** means the **frequent rules** that **also** pass the **minimum confidence**.

$$\text{Confidence}(X \Rightarrow Y) = \frac{|X \cup Y|}{|X|}$$

- For example frequent rules: {I1, I2} → {T100, T200, T300, T400, T500, T600, T700, T800, T900} with itemsets {1,2,5}, {2,4}, {2,3}, {1,2,4}, {1,3}, {2,3}, {1,3}, {1,2,3,5}, {1,2,3}
- Confidence(I1->I2)= 4/6 **pass 60%** (strong association rule!)
- Confidence(I2->I1)= 4/7



Strong Association Rule

- Confidence for 2 item set (1,2) (1,3) (1,5)

- $C(1 \rightarrow 2) = |1 \cup 2| / |1| = 4/6$

- $C(2 \rightarrow 1) = |1 \cup 2| / |2| = 4/7$

- $C(1 \rightarrow 3) = |1 \cup 3| / |1| = 4/6$

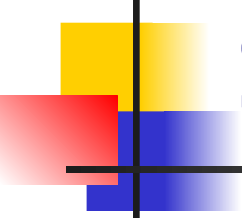
- $C(3 \rightarrow 1) = |1 \cup 3| / |3| = 4/6$

- $C(1 \rightarrow 5) = |1 \cup 5| / |1| = 2/6$

- $C(5 \rightarrow 1) = |1 \cup 5| / |5| = 2/2$ (100%)

$$Confidence(X \Rightarrow Y) = \frac{|X \cup Y|}{|X|}$$

■ T100	1,2,5
■ T200	2,4
■ T300	2,3
■ T400	1,2,4
■ T500	1,3
■ T600	2,3
■ T700	1,3
■ T800	1,2,3,5
■ T900	1,2,3



Strong Association Rule

- Confidence for 2 item set (2,3) (2,4) (2,5)

- $C(2 \rightarrow 3) = |2 \cup 3| / |2| = 4/7$

- $C(3 \rightarrow 2) = |2 \cup 3| / |3| = 4/6$

- $C(2 \rightarrow 4) = |2 \cup 4| / |2| = 4/7$

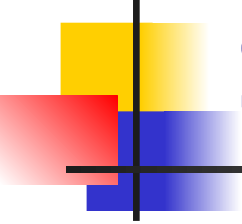
- $C(4 \rightarrow 2) = |2 \cup 4| / |4| = 2/2$ (100%)

- $C(2 \rightarrow 5) = |2 \cup 5| / |2| = 2/7$

- $C(5 \rightarrow 2) = |2 \cup 5| / |5| = 2/2$ (100%)

$$\text{Confidence}(X \Rightarrow Y) = \frac{|X \cup Y|}{|X|}$$

▪ T100	1,2,5
▪ T200	2,4
▪ T300	2,3
▪ T400	1,2,4
▪ T500	1,3
▪ T600	2,3
▪ T700	1,3
▪ T800	1,2,3,5
▪ T900	1,2,3



Strong Association Rule

- Confidence for 3 itemset (1, 2, 5)

- $C(1 \rightarrow 2, 5) = 2/6$

- $C(2 \rightarrow 1, 5) = 2/7$

- $C(5 \rightarrow 1, 2) = 2/2$

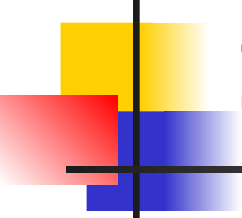
- $C(1, 2 \rightarrow 5) = 2/4$

- $C(1, 5 \rightarrow 2) = 2/2$

- $C(2, 5 \rightarrow 1) = 2/2$

$$\text{Confidence}(X \Rightarrow Y) = \frac{|X \cup Y|}{|X|}$$

■ T100	1,2,5
■ T200	2,4
■ T300	2,3
■ T400	1,2,4
■ T500	1,3
■ T600	2,3
■ T700	1,3
■ T800	1,2,3,5
■ T900	1,2,3



Strong Association Rule

- Confidence for 3 itemset (1, 2, 3)

- $C(1 \rightarrow 2, 3) = 2/6$

- $C(2 \rightarrow 1, 3) = 2/7$

- $C(3 \rightarrow 1, 2) = 2/6$

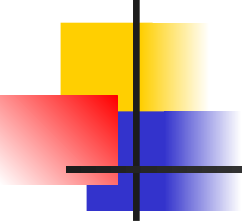
- $C(1, 2 \rightarrow 3) = 2/4$

- $C(1, 3 \rightarrow 2) = 2/4$

- $C(2, 3 \rightarrow 1) = 2/4$

$$\text{Confidence}(X \Rightarrow Y) = \frac{|X \cup Y|}{|X|}$$

■ T100	1,2,5
■ T200	2,4
■ T300	2,3
■ T400	1,2,4
■ T500	1,3
■ T600	2,3
■ T700	1,3
■ T800	1,2,3,5
■ T900	1,2,3

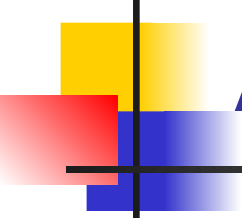


Exercise

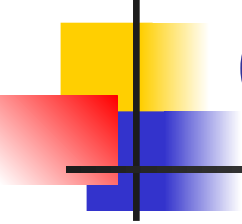
- A dataset has five transactions, let min-support=60% and min_confidence=80%
- Find all **strong association rules**

TID	Items_bought
T1	M, O, N, K, E,
T2	Y
T3	D, O, N, K , E,
T4	Y
T5	M, A, K, E
	M, U, C, K ,Y
	C, O, O, K, I ,E

Association Rules with Apriori

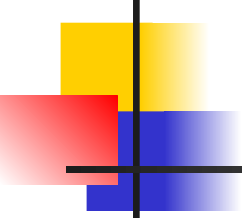


K:5		KE:4		<u>KE</u>	
E:4		KM:3		KM	
M:3		KO:3		<u>KO</u>	
O:3	=>	KY:3	=>	KY	=> KEO
Y:3		EM:2		<u>EO</u>	
		EO:3			
		EY:2			
		MO:1			
		MY:2			
		OY:2			



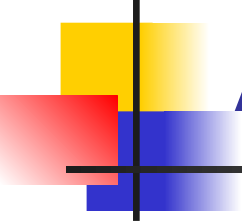
Outline

- Association Rules
- Misleading Rules
- Multi-level Association Rules



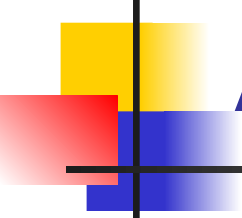
Example 5.8 Misleading “Strong” Association Rule

- Of the 10,000 transactions analyzed, the data show that
 - 6,000 of the customer included computer games,
 - while 7,500 include videos,
 - And 4,000 included both computer games and videos
 - $P(A \cup B) = 40\%$



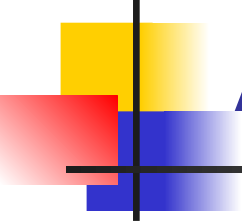
Misleading “Strong” Association Rule

- For this example:
 - $\text{Support (Game \& Video)} = 4,000 / 10,000 = 40\%$
 - $\text{Confidence (Game} \Rightarrow \text{Video)} = 4,000 / 6,000 = 66\%$
 - Suppose it pass our minimum support and confidence (30% , 60%, respectively)



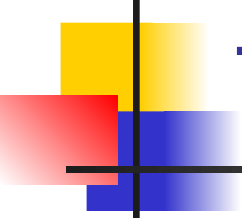
Misleading “Strong” Association Rule

- However, the truth is : “computer games and videos are negatively associated”
- Which means the purchase of one of these items actually decreases the likelihood of purchasing the other.
- (How to get this conclusion??)



Misleading “Strong” Association Rule

- Under the normal situation,
 - 60% of customers buy the game
 - 75% of customers buy the video
 - Therefore, it should have $60\% * 75\% = 45\%$ of people buy both
 - That equals to 4,500 which is more than 4,000 (the actual value)



From Association Analysis to Correlation Analysis

- **Lift** is a simple correlation measure that is given as follows
 - The occurrence of itemset A is independent of the occurrence of itemset B if
$$P(A \cup B) = P(A)P(B)$$
 - Otherwise, itemset A and B are **dependent** and **correlated** as events
- **$Lift(A, B) = P(A \cup B) / P(A)P(B)$**
 - If the value is less than 1, the occurrence of A is negatively correlated with the occurrence of B
 - If the value is greater than 1, then A and B are positively correlated
 - **$P(A \cup B) / P(A)P(B) = 0.4 / (0.6 * 0.75) = 0.4 / 0.45 < 1$**

Lift in our example

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☆

Strong Association Rule

- Confidence for 2 item set (1,2) (1,3) (1,5)

- $C(1 \rightarrow 2) = |1 \cup 2|/|1| = 4/6$
- $C(2 \rightarrow 1) = |1 \cup 2|/|2| = 4/7$
- $C(1 \rightarrow 3) = |1 \cup 3|/|1| = 4/6$
- $C(3 \rightarrow 1) = |1 \cup 3|/|3| = 4/6$
- $C(1 \rightarrow 5) = |1 \cup 5|/|1| = 2/6$
- $C(5 \rightarrow 1) = |1 \cup 5|/|5| = 2/2$ (100%)

$$\text{Confidence} = \frac{|P \cup Q|}{|P|}$$

- T100 1,2,5
- T200 2,4
- T300 2,3
- T400 1,2,4
- T500 1,3
- T600 2,3
- T700 1,3
- T800 1,2,3,5
- T900 1,2,3

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Strong Association Rule

- Confidence for 2 item set (2,3) (2,4) (2,5)

- $C(2 \rightarrow 3) = |2 \cup 3|/|2| = 4/7$
- $C(3 \rightarrow 2) = |2 \cup 3|/|3| = 4/6$
- $C(2 \rightarrow 4) = |2 \cup 4|/|2| = 4/7$
- $C(4 \rightarrow 2) = |2 \cup 4|/|4| = 2/2$ (100%)
- $C(2 \rightarrow 5) = |2 \cup 5|/|2| = 2/7$
- $C(5 \rightarrow 2) = |2 \cup 5|/|5| = 2/2$ (100%)

$$\text{Confidence} = \frac{|P \cup Q|}{|P|}$$

- T100 1,2,5
- T200 2,4
- T300 2,3
- T400 1,2,4
- T500 1,3
- T600 2,3
- T700 1,3
- T800 1,2,3,5
- T900 1,2,3

27

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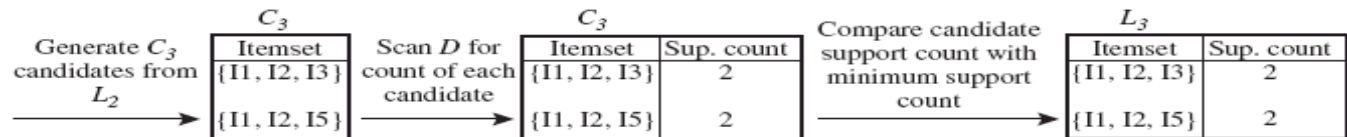
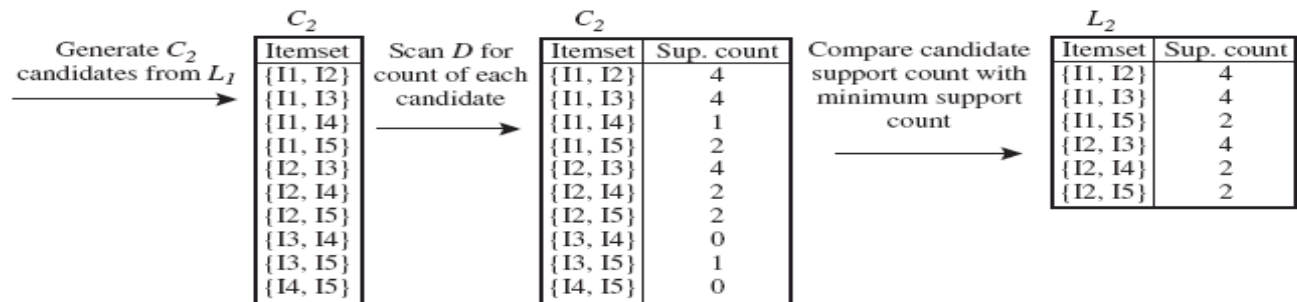
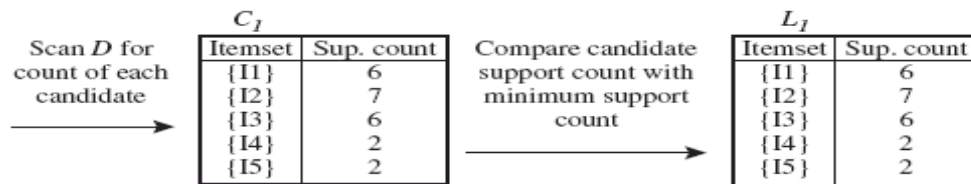
Strong Association Rule

- Confidence for 3 itemset (1, 2, 5)

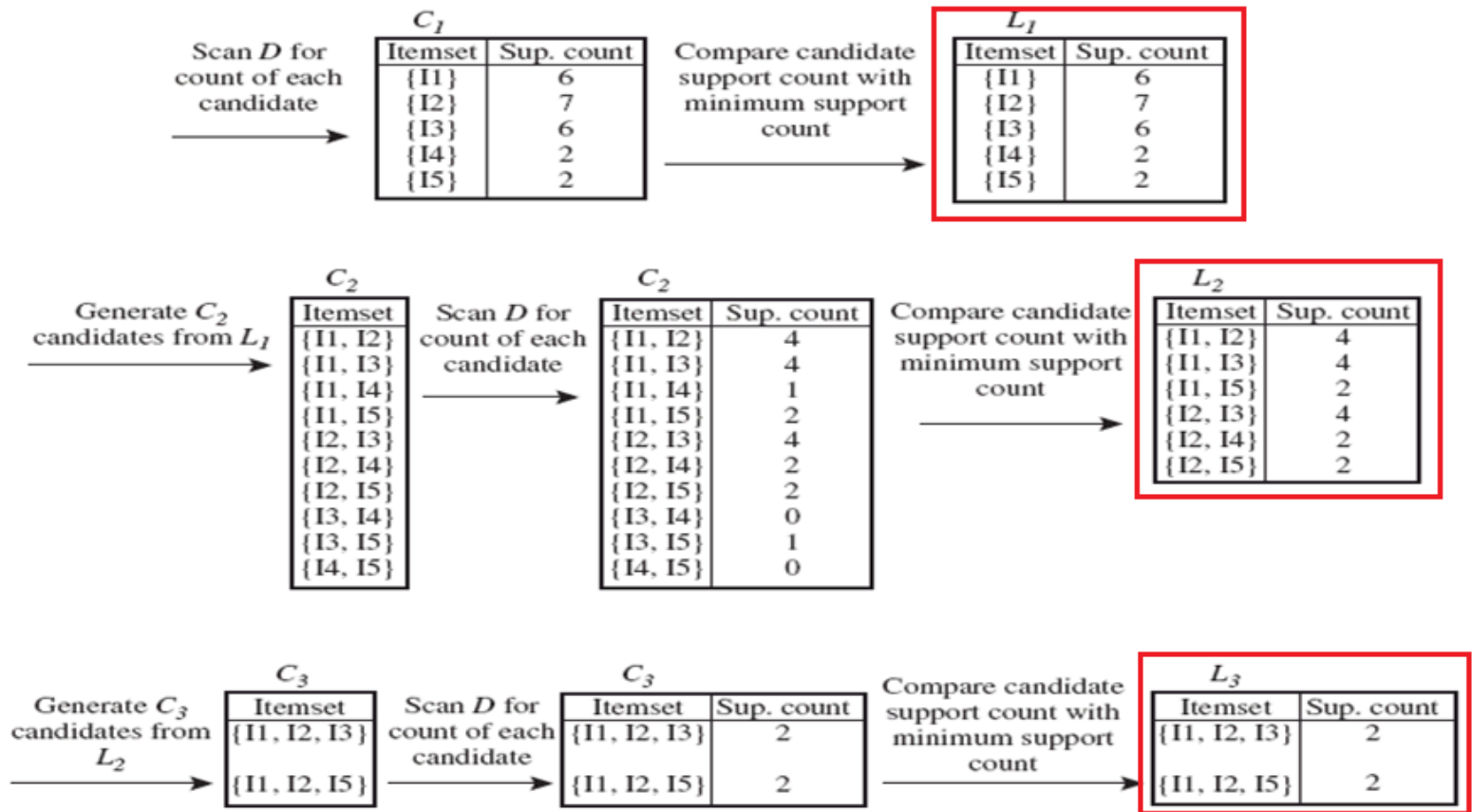
- $C(1 \rightarrow 2,5) = 2/6$
- $C(2 \rightarrow 1,5) = 2/7$
- $C(5 \rightarrow 1,2) = 2/2$
- $C(1,2 \rightarrow 5) = 2/4$
- $C(1,5 \rightarrow 2) = 2/2$
- $C(2,5 \rightarrow 1) = 2/2$

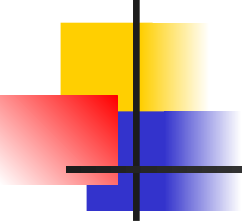
$$\text{Confidence} = \frac{|P \cup Q|}{|P|}$$

- T100 1,2,5
- T200 2,4
- T300 2,3
- T400 1,2,4
- T500 1,3
- T600 2,3
- T700 1,3
- T800 1,2,3,5
- T900 1,2,3



Lift in our example





Lift in our example

- $C(1 \rightarrow 2)$

$$\begin{aligned} & P(1,2)/P(1)*P(2) \\ &= (4/9) / (6/9)*(7/9) \\ &= 0.86 \end{aligned}$$

- $C(1 \rightarrow 3)$

$$\begin{aligned} & P(1,3)/P(1)*P(3) \\ &= (4/9) / (6/9)*(6/9) \\ &= 1.01 \end{aligned}$$

L_1

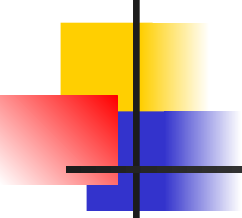
Itemset	Sup. count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

L_2

Itemset	Sup. count
{I1, I2}	4
{I1, I3}	4
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2

L_3

Itemset	Sup. count
{I1, I2, I3}	2
{I1, I2, I5}	2



Lift in our example

- $C(5 \rightarrow 1, 2)$

$$\begin{aligned} & P(5, 1, 2) / P(5) * P(1, 2) \\ &= (2/9) / (2/9) * (4/9) \\ &= 2.25 \end{aligned}$$

- $C(2, 5 \rightarrow 1)$

$$\begin{aligned} & P(2, 5, 1) / P(2, 5) * P(1) \\ &= (2/9) / (2/9) * (6/9) \\ &= 1.5 \end{aligned}$$

L_1

Itemset	Sup. count
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

L_2

Itemset	Sup. count
{I1, I2}	4
{I1, I3}	4
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2

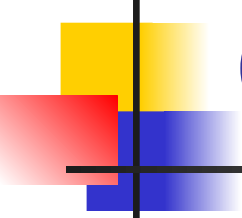
L_3

Itemset	Sup. count
{I1, I2, I3}	2
{I1, I2, I5}	2



From Association Analysis to Correlation Analysis

Strong Association Rules	Lift
C(1->2)	$P(1,2)/P(1)*P(2) = (4/9) / (6/9)*(7/9) = 0.86$
C(1->3)	$P(1,3)/P(1)*P(3) = (4/9) / (6/9)*(6/9) = 1.01$
C(3->1)	$P(3,1)/P(3)*P(1) = (4/9) / (6/9)*(6/9) = 1.01$
C(5->1)	$P(5,1)/P(5)*P(1) = (2/9) / (2/9)*(6/9) = 1.49$
C(3->2)	$P(3,2)/P(3)*P(2) = (4/9) / (6/9)*(7/9) = 0.86$
C(4->2)	$P(4,2)/P(4)*P(2) = (2/9) / (2/9)*(7/9) = 1.29$
C(5->2)	$P(5,2)/P(5)*P(2) = (2/9) / (2/9)*(7/9) = 1.29$
C(5->1,2)	$P(5,1,2)/P(5)*P(1,2) = (2/9) / (2/9)*(4/9) = 2.25$
C(1,5->2)	$P(1,5,2)/P(1,5)*P(2) = (2/9) / (2/9)*(7/9) = 1.29$
C(2,5->1)	$P(2,5,1)/P(2,5)*P(1) = (2/9) / (2/9)*(6/9) = 1.5$

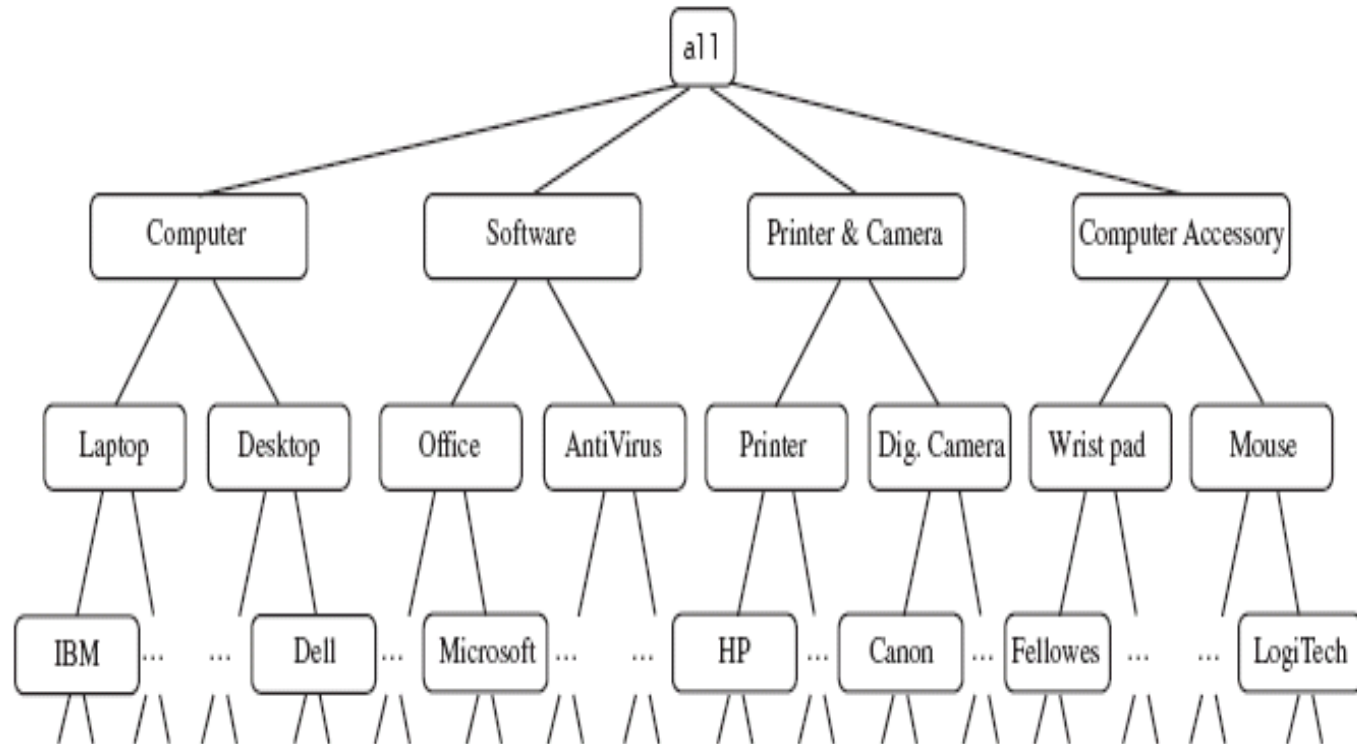


Outline

- Association Rules
- Misleading Rules
- Multi-level Association Rules

Mining Multiple-Level Association Rules

- Items often form hierarchies



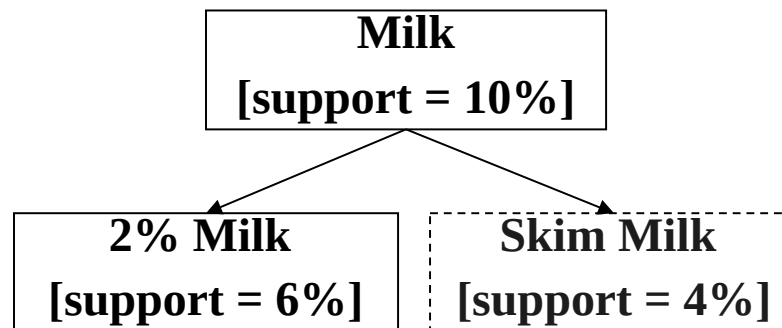
Mining Multiple-Level Association Rules

- Flexible support settings
 - Items at the lower level are expected to have lower support

uniform support

Level 1
min_sup = 5%

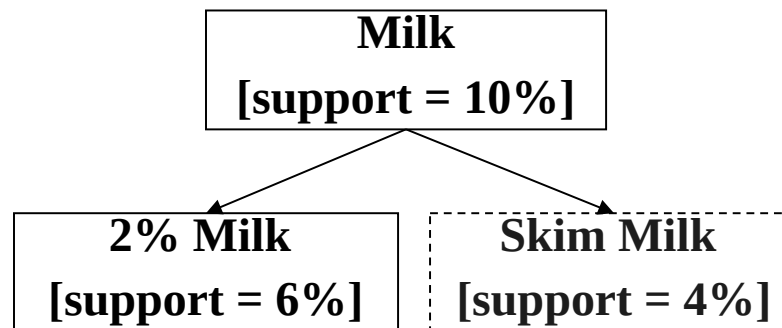
Level 2
min_sup = 5%

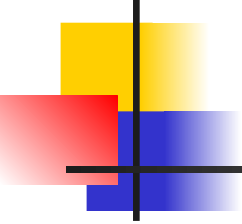


reduced support

Level 1
min_sup = 5%

Level 2
min_sup = 3%





Multi-level Association: Redundancy Filtering

- Some rules may be redundant due to “ancestor” relationships between items.
- Example
 - milk $\Rightarrow$ wheat bread [support = 8%, confidence = 70%]
 - 2% milk $\Rightarrow$ wheat bread [support = 2%, confidence = 72%]
- We say the first rule is an ancestor of the second rule.